

Security+ 3.4 · Capacity Planning

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.4** —
Capacity Planning (3:53) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Capacity planning ensures that people, technology, and infrastructure can meet current and future demand. Key metrics: MTTR (Mean Time To Repair) measures response time; MTBF (Mean Time Between Failures) measures reliability —how infrequently failures occur.

High-Yield Core Concepts Tested on Exam

- Capacity = trained personnel + hardware + infrastructure
- MTBF = Mean Time Between Failures (reliability metric)
- MTTR = Mean Time To Repair (response time metric)
- Capacity planning aligns resources with demand
- MTBF indicates how often failures occur; higher is better
- MTTR indicates how quickly systems are restored; lower is better

Exam Tips & Classic Scenarios

- 'mean time between failures' = MTBF
- 'mean time to repair' = MTTR
- MTBF measures reliability; MTTR measures responsiveness

Typical Exam Scenario: A server fails once every 1000 hours. This indicates an MTBF of 1000 hours. If it takes 2 hours to repair, the MTTR is 2 hours.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.4 — Capacity Planning)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. RTO (Recovery Time Objective) measures:

- A)** How much DATA you accept losing
- B)** How much TIME the service can be down
- C)** The cost of the incident
- D)** The number of affected users

2. If a server fails and must be operational within 4 hours according to an SLA, this defines:

- A) RPO (Recovery Point Objective)**
- B) RTO (Recovery Time Objective)**
- C) MTTR (Mean Time To Repair)**
- D) MTBF (Mean Time Between Failures)**

3. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Capacity = trained people + hardware + infrastructure'. What is the best recommendation or applicable control classification?

- A) Implement the corresponding solution: Capacity = trained people + hardware + infrastructure...**
- B) Temporarily disable audit logs to accelerate system throughput**
- C) Replace preventive technical controls with informal terms of adherence**
- D) Allow shared access among administrators with generic credentials**

4. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'MTBF = mean time between failures (reliability); MTTR = mean time to repair'. What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Implement the corresponding solution: MTBF = mean time between failures (reliability); MTTR = mean time to repair...**
- C) Replace preventive technical controls with informal terms of adherence**
- D) Allow shared access among administrators with generic credentials**

5. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'mean time between failures' = MTBF. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'mean time between failures' = MTBF.
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

6. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: "server fails once every 1000h'=MTBF of 1000h.'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'server fails once every 1000h'=MTBF of 1000h....
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Capacity = trained people + hardware + infrastructure'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Capacity = trained people + hardware + infrastructure...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'MTBF = mean time between failures (reliability); MTTR = mean time to repair'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: MTBF = mean time between failures (reliability); MTTR = mean time to repair...
- B)** Temporarily disable audit logs to speed up system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'mean time between failures'=MTBF. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'mean time between failures'=MTBF...

10. Corporate scenario (Capacity Planning): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: '"server fails once every 1000h'=MTBF of 1000h'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'server fails once every 1000h'=MTBF of 1000h....
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — How much TIME the service can be down.

✓ **Why it's correct:** The correct answer is: how much TIME the service can be down. RTO (Recovery Time Objective) is a central metric in the Business Continuity Plan (BCP) and Disaster Recovery Plan (DRP), covered in the Business Continuity and Disaster Recovery domain of CompTIA Security+ SY0-701. It defines the maximum acceptable time to restore a service or process after an interruption, before the business impact becomes unacceptable. For example, if an e-commerce site sets an RTO of 2 hours, the recovery team must restore operations within 2 hours after the incident. RTO guides the choice of recovery strategies (hot site, warm site, cold site), backup frequency, and investments in redundancy. It is often confused with RPO, but while RTO measures downtime, RPO measures data loss.

✗ **Why the others are wrong:**

- *How much DATA you accept losing:* This is the definition of RPO (Recovery Point Objective), not RTO. RPO establishes the maximum amount of data that can be lost, measured in time (e.g., 15 minutes of transactions). It determines backup frequency: if the RPO is 1 hour, backups must occur at least every hour. RTO, on the other hand, focuses on service restoration time, not data loss. Confusing the two concepts leads to inadequate recovery strategies, as a low RPO does not guarantee a low RTO, and vice versa.
- *The cost of the incident:* The cost of the incident is a financial metric that quantifies direct and indirect losses resulting from an interruption (lost revenue, fines, productivity, reputational damage). Although it is an input to justify investments in continuity, it is not the definition of RTO. RTO is a time goal, not a monetary value. Using cost as the definition of RTO would divert focus from recovery planning, which needs clear temporal parameters to guide technical actions.
- *The number of affected users:* The number of affected users is an impact metric, useful for prioritization and communication, but it does not define RTO. RTO is a recovery time objective, regardless of the number of users. A service critical to few users may have a short RTO, while a service for many users may have a longer RTO, depending on business criticality. Therefore, this alternative does not correspond to the formal concept of RTO.

2) B — RTO (Recovery Time Objective).

✓ **Why it's correct:** RTO (Recovery Time Objective) is the maximum acceptable time to restore a service or system after a disruption, as defined in business continuity objectives. If an SLA states that the server must be operational within 4 hours, that is exactly the RTO: the target time for recovery. CompTIA Security+ distinguishes RTO from RPO and other metrics, with RTO focusing on tolerable downtime, not data loss.

✗ **Why the others are wrong:**

- *RPO (Recovery Point Objective):* RPO (Recovery Point Objective) defines the maximum amount of data the organization can lose, measured in time (e.g., 1 hour of transactions). It relates to backup frequency and tolerance for data loss, not service recovery time. A 4-hour SLA to restore the server does not describe data loss but rather the maximum downtime, so it is not RPO.
- *MTTR (Mean Time To Repair):* MTTR (Mean Time To Repair) is a statistical metric representing the average time required to repair a component or system after a failure. It is a historical and operational measure, not a contractual objective or recovery goal. The 4-hour SLA is a prescriptive recovery target, not an average of past repairs. Confusing MTTR with RTO is a common conceptual error but technically distinct.
- *MTBF (Mean Time Between Failures):* MTBF (Mean Time Between Failures) is the average time between consecutive failures of a repairable system, indicating its reliability. It does not establish

recovery deadlines but rather the expected frequency of failures. A 4-hour SLA for restoration is unrelated to failure frequency but to recovery speed. Therefore, MTBF does not answer the given definition.

3) A — Implement the corresponding solution: Capacity = trained people + hardware + infrastructure....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Capacity Planning): Capacity = trained people + hardware + infrastructure.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) B — Implement the corresponding solution: MTBF = mean time between failures (reliability); MTTR = mean time to repair....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Capacity Planning): MTBF = mean time between failures (reliability); MTTR = mean time to repair.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) A — Implement the corresponding solution: 'mean time between failures' = MTBF..

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Capacity Planning): 'mean time between failures' = MTBF. MTBF is a reliability metric that measures the average time between failures of a system or component. It is used in capacity planning and availability analysis to predict system uptime and plan maintenance. Security+ covers MTBF, MTTR, and other metrics as part of resilience and recovery objectives.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaches compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.

- *Allow shared access among administrators with generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: 'server fails once every 1000h'=MTBF of 1000h.....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Capacity Planning): 'server fails once every 1000h'=MTBF of 1000h. Mean Time Between Failures (MTBF) is the predicted elapsed time between inherent failures of a system during operation.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) B — Implement the corresponding solution: Capacity = trained people + hardware + infrastructure....

✓ **Why it's correct:** According to the official CompTIA Security+ exam objectives (SY0-701) for lesson 3.4 (Capacity Planning): Capacity = trained people + hardware + infrastructure.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who made critical changes or malicious commands.

8) A — Implement the corresponding solution: MTBF = mean time between failures (reliability); MTTR = mean time to repair....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Capacity Planning): MTBF = mean time between failures (reliability); MTTR = mean time to repair. MTBF measures the average time between failures of a system or component, indicating reliability. MTTR measures the average time to repair a failed component, indicating maintainability. These metrics are critical for capacity planning, availability, and resilience. The appropriate control is to implement the corresponding solution, which involves monitoring these metrics and designing systems for high availability.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to speed up system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks

compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: 'mean time between failures'=MTBF....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Capacity Planning): 'mean time between failures'=MTBF. MTBF is a reliability metric that measures the average time between failures of a system or component, used in capacity planning to predict availability and plan maintenance.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) A — Implement the corresponding solution: 'server fails once every 1000h'=MTBF of 1000h.....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Capacity Planning): 'server fails once every 1000h'=MTBF of 1000h. MTBF (Mean Time Between Failures) is a reliability metric that indicates the average time between failures of a repairable system. It is used in capacity planning and resiliency to predict failure rates and plan maintenance or redundancy.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

